

CV Assist: A Real-Time Visual Assistance System for the Visually Impaired

Empowering Independence Through Open-Vocabulary Detection, Monocular Depth, Hand Tracking & LLM-Vision

Developed by [Your Name / Team]

OWL-VIT • MIDAS • MEDIAPIPE • DEEPSEEK LLM-VISION

Visually Impaired Users Lack Intuitive, Real-Time Spatial Guidance

THE CHALLENGE

Over 2.2 billion people live with vision impairment. Daily tasks like object localization and grasping remain major barriers to independence.

CURRENT GAP

Existing tools lack real-time spatial guidance. Current AI solutions often require specialized hardware or fixed categories, limiting their utility.

THE OPPORTUNITY

Integrating open-vocabulary search with depth-aware spatial guidance and natural voice interaction into a single, accessible pipeline.

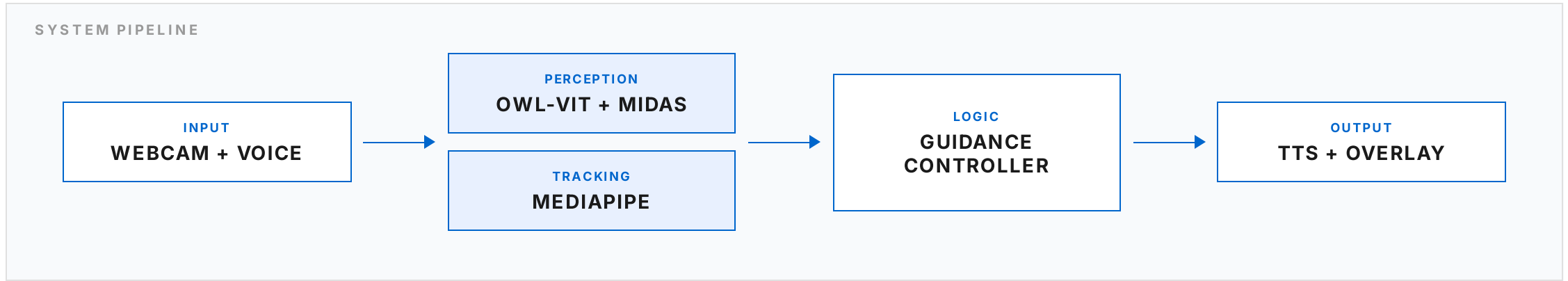
OUR GOAL

A **"Virtual Eye"** that guides the user's hand to any object using natural language and real-time audio feedback.



Bridging the gap between **perception** and **action** through intuitive guidance.

Four Decoupled Modules Work in Concert for Real-Time Guidance



■ Modular Pipeline

Decoupled perception modules (OWL-ViT, MiDaS, MediaPipe) run in parallel, allowing for independent upgrades and optimizations.

■ Multimodal Input

Fuses continuous RGB video streams with asynchronous voice commands to create a flexible, hands-free interaction model.

■ Real-Time Output

The controller synthesizes spatial data to deliver immediate, actionable audio instructions and visual feedback.

Search for Any Object by Name — No Retraining Required

■ TECHNOLOGY: OWL-VIT V2

Utilizes the Open-Vocabulary Vision Transformer, a CLIP-based detector that accepts free-text queries instead of relying on predefined classes.

■ ZERO-SHOT CAPABILITY

Detects any object category described in natural language (e.g., "red coffee cup", "TV remote") without requiring model fine-tuning or retraining.

■ BOUNDING BOX OUTPUT

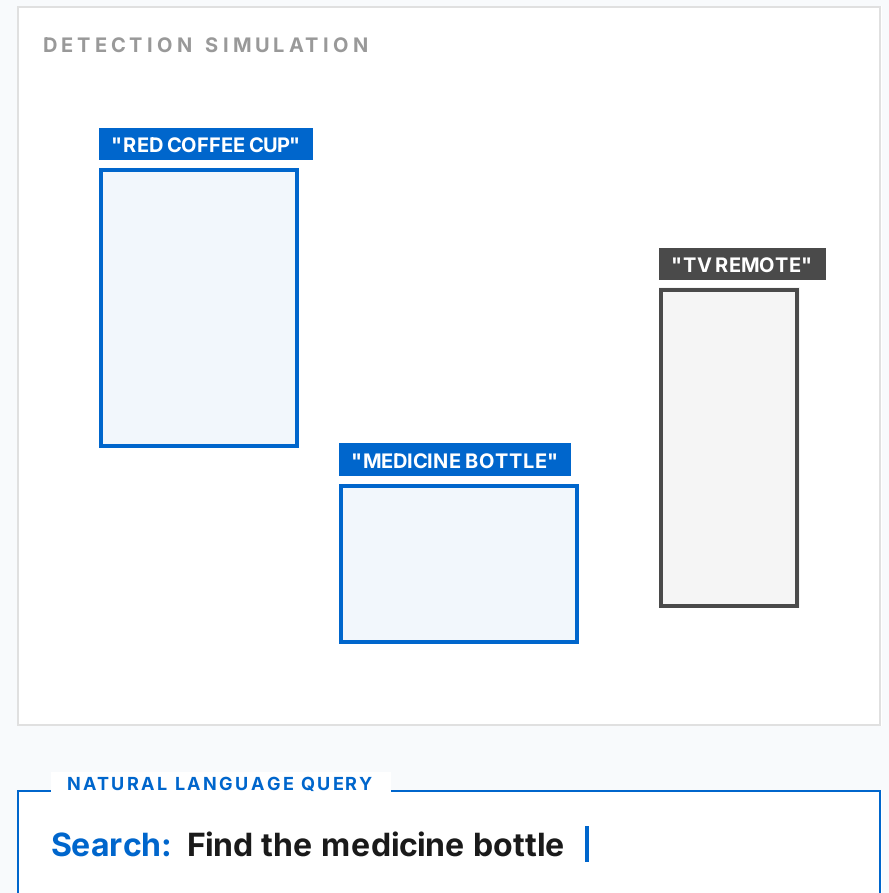
Provides precise 2D object location coordinates, serving as the foundational input for downstream spatial guidance and depth computation.

■ ROBUSTNESS

Maintains reliable detection accuracy across diverse lighting conditions, varying angles, and highly cluttered real-world environments.

■ THE IMPACT

Eliminates the rigid category constraints of traditional detectors, making the system universally applicable to any user's specific needs.



Accurate Distance Sensing from a Single Consumer Camera

□ TECHNOLOGY

MiDaS_small — a lightweight monocular depth estimation model optimized for real-time inference on standard hardware.

□ FUNCTION

Generates a per-pixel relative depth map from a single RGB frame, estimating distance to both the target and the user's hand.

□ KEY ADVANTAGE

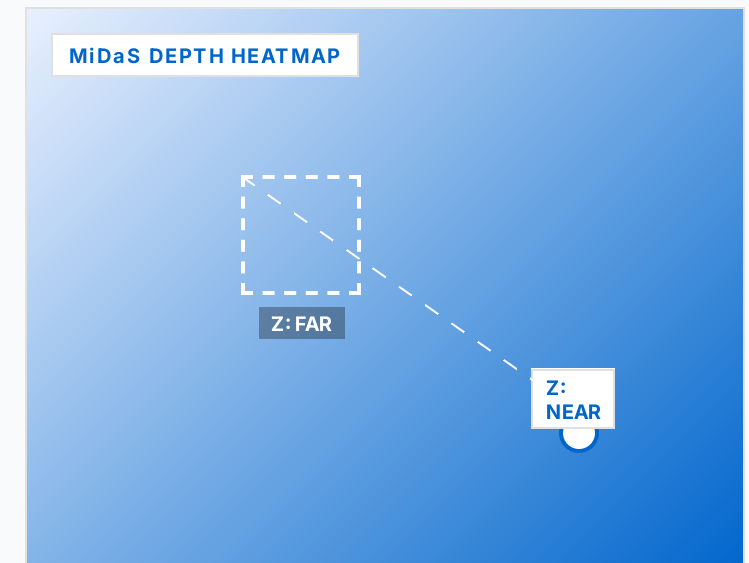
Eliminates the need for expensive stereo cameras or LiDAR — runs entirely on any standard consumer-grade webcam.

□ INTEGRATION

Depth values are sampled at the object center and hand landmark to compute the **Z-axis** guidance component.

□ PERFORMANCE

Optimized inference ensures zero lag in spatial perception, maintaining high frame rates for fluid interaction.



Computing **Z-axis offset** between hand and target using relative depth estimation.

High-Fidelity 3D Hand Landmarks Enable Precise Spatial Alignment

TECHNOLOGY

Google MediaPipe Hands tracks 21 3D hand landmarks in real time, providing robust skeletal tracking from a standard RGB feed.

HAND POSITION

The centroid of key landmarks (wrist + finger bases) is calculated as the definitive "hand location" for spatial offset computation.

GESTURE STATES

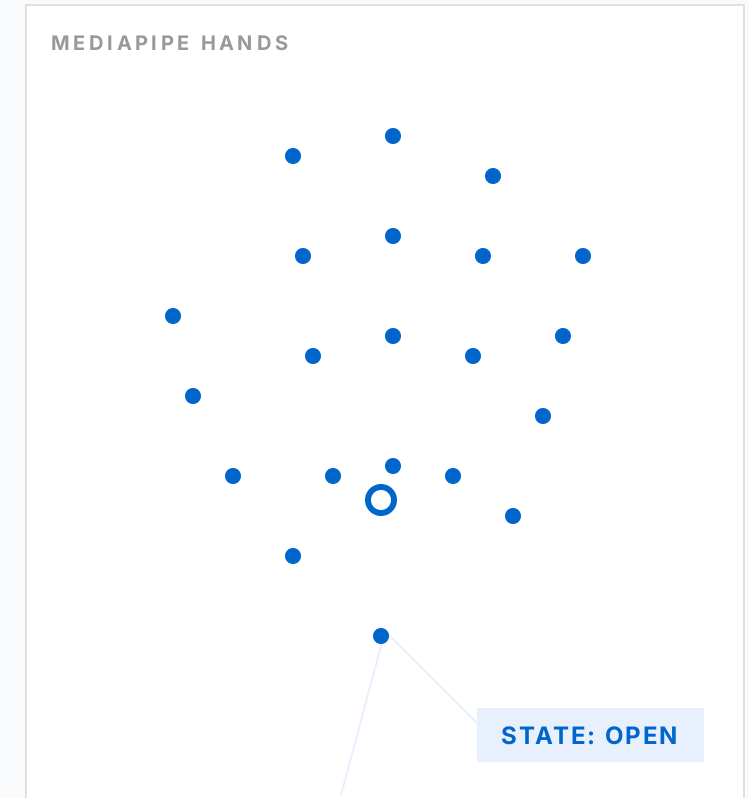
- **OPEN HAND:** Reaching / Ready to grasp.
- **CLOSED FIST:** Object grabbed; task complete.

SPATIAL ALIGNMENT

The X-Y offset between the hand centroid and the object center drives the directional audio instructions.

PRECISION

Sub-centimeter accuracy enables fine-grained, highly responsive guidance as the hand approaches the target.



Real-time tracking of **21 3D landmarks** for precise hand-to-object alignment.

Dual-Threshold Logic Delivers Stable, Professional-Grade Audio Instructions

THE PROBLEM: INSTRUCTION JITTER

Naive threshold-based guidance causes rapid oscillation ("Left... Right... Left...") when the hand is near the alignment boundary, making the system confusing.

THE SOLUTION: HYSTERESIS ALGORITHM

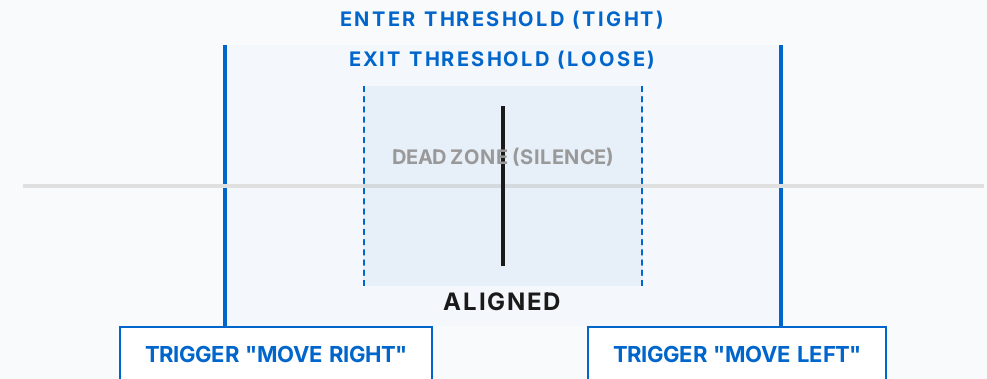
Implements two distinct thresholds for state transitions:

Enter Threshold (Outer): A new directional instruction is triggered only when the offset exceeds a larger boundary.

Exit Threshold (Inner): The instruction is cancelled only when the offset drops below a smaller boundary.

THE RESULT: STABLE GUIDANCE

Creates a stable "**dead zone**" around alignment. Instructions are issued decisively and cancelled cleanly, ensuring calm and confident spatial audio feedback.



1D Spatial Offset Axis demonstrating dual-threshold state transitions for **jitter-free** interaction.

DeepSeek-v3 Uses Live Camera Frames to Fix Voice Recognition Errors

THE PROBLEM

ASR (Whisper) frequently mishears object names due to accents or noise (e.g., "cut" instead of "cup").

MULTIMODAL SOLUTION

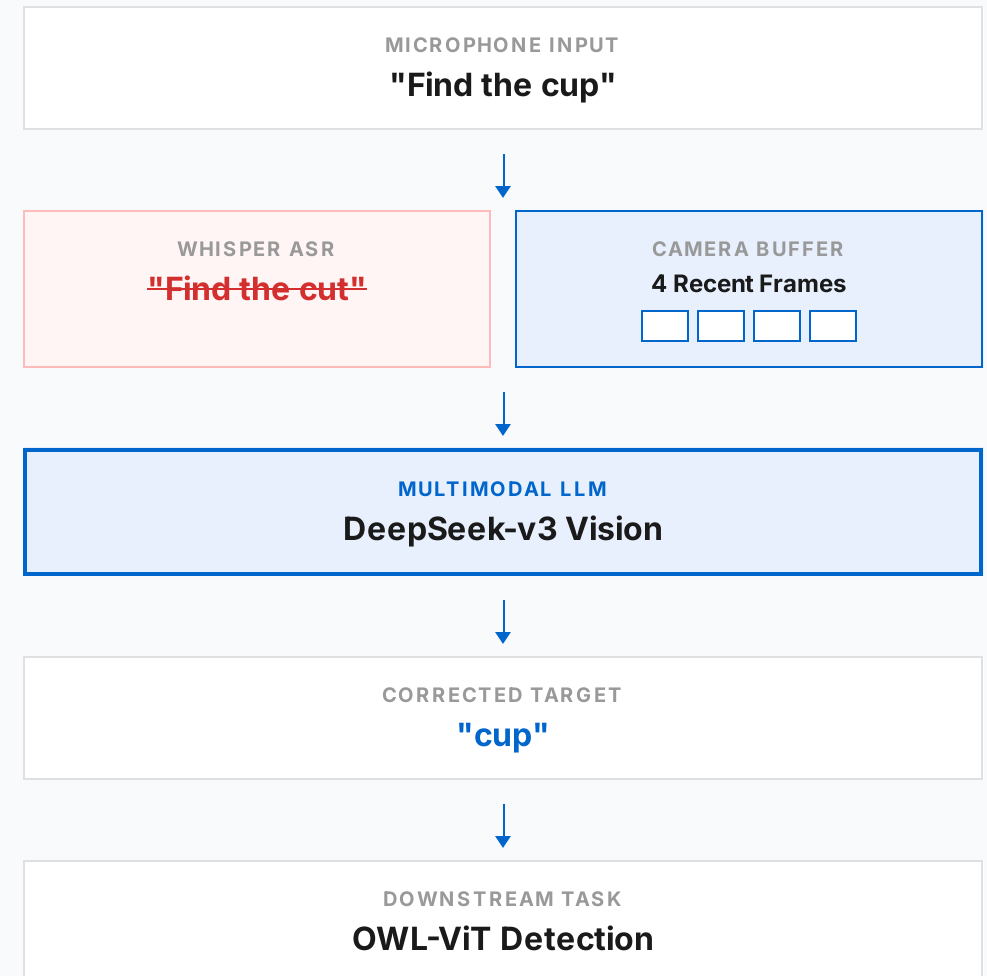
Whisper transcribes the command; the last 4 camera frames are encoded. DeepSeek-v3 uses both to return the most likely object name based on visual context.

ROBUSTNESS

Handles accents and phonetically similar object names by grounding audio in the visual scene.

ASYNCHRONOUS EXECUTION

LLM correction runs in a background thread, ensuring it does not block the real-time CV pipeline.



Real-Time Performance on Consumer Hardware — No Specialized Sensors Needed

■ FP16 INFERENCE

All models run in FP16, halving memory use and boosting throughput.

■ INTELLIGENT FRAME SKIPPING

Detection runs every N frames; depth and hand tracking run each frame to balance accuracy and speed.

■ ASYNCHRONOUS AUDIO

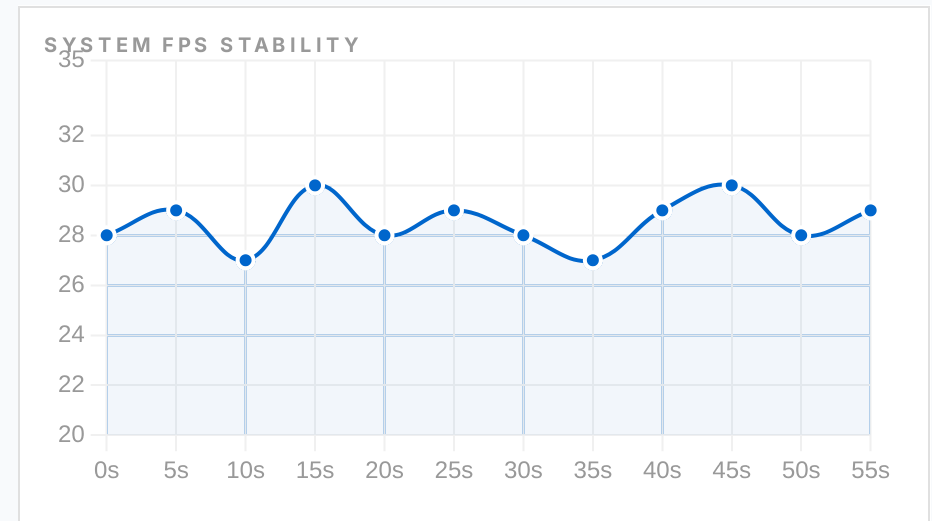
TTS runs in a background thread so audio never blocks the video processing loop.

■ CONSUMER HARDWARE

Stable 25–30 FPS on mid-range GPUs (e.g., NVIDIA RTX 3060) without LiDAR or special sensors.

■ LOW LATENCY

End-to-end latency is under 200ms for intuitive, real-time feedback.



28.5

AVERAGE FPS

<200_{ms}

END-TO-END LATENCY

CV Assist Demonstrates That Consumer AI Can Meaningfully Restore Independence

■ IMPACT

CV Assist provides visually impaired users with a practical, low-cost "virtual eye" for independent object grasping — a task previously requiring sighted assistance.

■ TECHNICAL ACHIEVEMENT

Successfully integrates four state-of-the-art AI modules (OWL-ViT, MiDaS, MediaPipe, DeepSeek LLM-Vision) into a coherent, real-time pipeline.

■ KEY INNOVATIONS

- Zero-shot open-vocabulary object search
- Hysteresis-based jitter-free spatial guidance
- LLM-Vision multimodal ASR error correction

■ FUTURE DIRECTIONS

- Expand gesture vocabulary for richer interaction.
- Deploy on edge devices (Raspberry Pi 5, NVIDIA Jetson).
- Integrate haptic feedback (vibration wristband).

SYSTEM VISION

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**Seeing the world,
one object at a time.**

A FUTURE WHERE AI-POWERED COMPUTER
VISION IS A STANDARD ACCESSIBILITY
LAYER.